

RouteX India

AI-Powered Digital Freight Marketplace & Intelligent Fleet Management System

Comprehensive Technical Architecture & System Specification Report

Document Version: 2.0 (Verified Monorepo Audit)
Primary Target OS: Linux / Windows / Cloud Kubernetes Environment
Backend Core: NestJS 11 + Prisma ORM 5 + PostgreSQL (Supabase)
Frontend Web: Next.js 16 (App Router) + React 19 + Tailwind CSS v4
Mobile Apps: React Native & Expo (Android)
AI Engine: Groq Cloud SDK (Llama 3.3 70B Versatile)
GIS & Telemetry: Leaflet, OSRM Routing, Socket.IO WebSockets
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EXECUTIVE SUMMARY:
RouteX India is an enterprise digital freight platform designed to automate commercial road transportation across India. It solves key logistics challenges—opaque broker pricing, manual phone dispatch, lack of shipment visibility, and paper POD delays—by combining real-time OSRM GIS routing, Groq Llama 3.3 dynamic freight pricing, Socket.IO live GPS telemetry, multi-stop delivery management, electronic Proof of Delivery (ePOD), and automated digital wallet escrow settlements.

1. ROUTEX — SYSTEM OVERVIEW & BUSINESS ARCHITECTURE

1.1 Core Mission & Industry Context

RouteX India addresses systemic inefficiencies in the Indian road logistics sector. By digitizing freight transactions between Shippers, Transporters (Fleet Owners), and Drivers, RouteX replaces manual broker phone calls with automated algorithmic matching, real-time GIS route calculation, and instant digital payments.

1.2 Target User Personas & Ecosystem

- Shippers (Companies & Individuals):** Book commercial freight, receive instant AI pricing quotes, monitor live GPS movement, and receive verified digital PODs.
- Transporters (Fleet Owners):** Manage vehicle fleets, assign drivers to loads, monitor maintenance/fuel/tyres, and receive automated wallet settlements.
- Drivers (Owner-Operators & Employees):** Receive load dispatch alerts via mobile app, stream live GPS telemetry, execute multi-stop deliveries, and capture digital signatures.
- Super Admins:** Monitor national freight dispatch, review AI pricing telemetry, verify user credentials, and resolve platform disputes.

2. REPOSITORY TOPOLOGY & APPS SPECIFICATION

Application	Directory	Tech Stack	Role & Core Responsibility
routeX-backend	routeX-backend/	NestJS 11, Prisma 5, PostgreSQL, Winston, Socket.IO	Core REST API engine, Groq AI pricing, WebSocket tracking, Razorpay gateway, FCM notifications.
routeX-web	routeX-web/	Next.js 16 (App Router), React 19, Tailwind CSS v4, Leaflet	Multi-role web portal for Shippers, Fleet Owners, and Super Admins.
routeX-driver	routeX-driver/	React Native, Expo, Android SDK	Driver mobile app for load acceptance, live GPS streaming, and ePOD upload.
routeX-mobile	routeX-mobile/	React Native, Expo	Customer mobile application for booking and tracking on mobile devices.
Infrastructure	Root (/), k8s/	Docker Compose, Kubernetes, Nginx	Production container orchestration, SSL proxying, and load balancing.

3. CUSTOM AUTHENTICATION & SESSION ARCHITECTURE

- RouteX utilizes a **production-ready custom authentication engine** fully decoupled from third-party auth vendors like Clerk.
- Primary Method:** Phone Number + 6-Digit OTP SMS.
 - Phone Normalization:** Normalizes Indian mobile numbers (e.g. 9876543210, +91 9876543210) to standard E.164 (+919876543210) using strict regex validation `/^\+91[6-9]\d{9}$/`.
 - Security Controls:** SHA-256 OTP hashing (plain text OTP is never persisted in database), 5-minute strict expiry, 60-second resend cooldown, max 3 verification attempts per OTP code, and immediate single-use invalidation (`verified_at`).
 - Token Rotation Engine:** Issues 7-day JWT access tokens and 30-day SHA-256 hashed refresh tokens stored in the `auth_sessions` database table. Endpoint `POST /api/auth/refresh` performs token rotation.
 - Multi-Provider Abstraction (ISmsProvider):** Supports Fast2SMS (Indian OTP gateway), Twilio (Global SMS), MSG91 (Enterprise India), and Console Mock Provider (development mode logging).

4. AI DYNAMIC PRICING ENGINE & GROQ LLM INTEGRATION

- Freight rate calculation is powered by an advanced dual-engine system located in `routex-backend/src/ai/pricing.service.ts`:
- 1. **Groq LLM Engine:** Queries Groq Cloud SDK using the `llama-3.3-70b-versatile` model. The service supplies distance, tonnage, truck class, fuel pricing, weather conditions, and traffic level, instructing the LLM to output a strictly formatted JSON object containing estimated price, confidence score, and cost breakdown.
 - 2. **Deterministic Local Rule Engine:** If Groq API key is unconfigured or network failure occurs, the service automatically fails over to a local formula engine calculating:

$$\text{Price} = \text{BasePrice} + (\text{Distance_Km} * \text{PerKmRate}) * \text{WeightFactor} * \text{WeatherSurcharge} * \text{TrafficSurcharge} + \text{TollEstimate}$$

5. LIVE GPS TRACKING & SOCKET.IO WEBSOCKET ENGINE

Event Name	Sender	Target Room	Payload Data Structure	Functional Purpose
driver:join	Driver App	Server	{ driverId }	Registers driver socket mapping for targeted dispatching.
shipper:join	Shipper Web	Server	{ bookingId }	Subscribes shipper socket to live tracking room for a trip.
driver:locationUpdate	Driver App	Shipper, Fleet, Admin	{ driverId, bookingId, latitude, longitude, speed, heading }	Streams real-time GPS telemetry to animate map markers.
booking:delay_detected	Server	Shipper, Admin	{ bookingId, stopOrder, durationMinutes, message }	Emits alert when vehicle stays stationary > 15 mins.
booking:completed	Server	Shipper, Fleet	{ bookingId, status: 'completed' }	Triggers trip completion and releases wallet funds.

6. COMPLETE BUTTON & ACTION INVENTORY

Button Name	Location / Screen	Role	API Endpoint Called	Backend & Database Effect
Send Verification OTP	/login, /register	Public	POST /api/auth/phone/send-otp	Generates 6-digit OTP, stores SHA-256 hash in phone_verifications, sends SMS.
Verify OTP & Access	/login, /register	Public	POST /api/auth/phone/verify-otp	Validates OTP hash, creates user/profile, issues JWT session token.
Calculate Instant Rate	/shipper/book	Shipper	POST /api/ai/pricing	Queries OSRM GIS distance and Groq Llama 3.3 LLM for pricing breakdown.
Confirm & Book Truck	/shipper/book	Shipper	POST /api/bookings	Inserts Booking + booking_stops, emits Socket & FCM alerts.
Accept Load	Driver App	Driver	POST /api/bookings/:id/accept	Assigns driver to booking, updates status to assigned.
Arrived at Pickup	Driver App	Driver	PUT /api/bookings/:id/status	Updates status to at_pickup, broadcasts socket status update.
Start Trip	Driver App	Driver	PUT /api/bookings/:id/status	Updates status to in_transit, initiates live GPS streaming.
Submit ePOD & Complete	Driver App	Driver	POST /api/bookings/:id/epod	Stores signature/photo URL, completes trip, transfers wallet funds.
Add Truck	/fleet/vehicles	Transporter	POST /api/fleet/trucks	Inserts new record into trucks table under company ID.
Assign Driver	/fleet/dispatch	Transporter	POST /api/fleet/assign-driver	Pairs driver_id to trucks.driver_id.
Request Payout	/fleet/wallet	Transporter	POST /api/fleet/wallet/payout	Deducts wallet balance, inserts bank transfer payout transaction.

7. DATABASE SCHEMA & ENTITY MODEL (PRISMA / SUPABASE)

Model / Table	Primary Key	Key Foreign Keys	Purpose & Business Meaning
users & profiles	id (UUID)	id $\rightarrow$ users.id	User identity, name, role (shipper, fleet_owner, driver, super_admin), phone.
phone_verifications	id (UUID)	N/A	Stores SHA-256 hashed 6-digit OTPs, attempts, 5m expiry, 60s resend cooldown.
auth_sessions	id (UUID)	user_id	Tracks active JWT refresh token hashes, device IDs, and revocation status.
Booking	id (UUID)	shipper_id, driver_id, truck_id	Freight order record, status state machine, quoted price, signature URLs.
booking_stops	id (UUID)	booking_id	Multi-stop waypoints, sequence order, lat/lng, per-stop ePOD signature.
trucks & drivers	id (UUID)	company_id, driver_id	Commercial vehicle fleet registry, driver license verification, and availability.
wallets & transactions	id (UUID)	profile_id	Double-entry digital wallet ledger for deposits, escrow holds, earnings, and payouts.
push_tokens	id (UUID)	user_id	FCM device push tokens for multi-platform notifications (android, ios, web).

8. SECOND-PASS VERIFIED CODEBASE TRUTH TABLE

The following truth table represents the verified code-level audit of all major RouteX platform capabilities:

Feature / Capability	Status Classification	Implementation Source File	API / Gateway Route	Technical Audit Notes
Phone OTP Auth	■ FULLY IMPLEMENTED	src/auth/sms.service.ts	POST /api/auth/phone/send-otp	SHA-256 hashed OTPs, 60s cooldown, 5m expiry.
Token Rotation Sessions	■ FULLY IMPLEMENTED	src/auth/auth.service.ts	POST /api/auth/refresh	Stores refresh hashes in auth_sessions table.
Multi-Role RBAC	■ FULLY IMPLEMENTED	src/auth/roles.guard.ts	Guard @Roles()	Enforces shipper, fleet_owner, driver, admin.
AI Freight Pricing	■ FULLY IMPLEMENTED	src/ai/pricing.service.ts	POST /api/ai/pricing	Groq SDK (Llama 3.3 70B) + local formula fallback.
AI Route Optimization	■ MOCK / PLACEHOLDER	src/map/map.service.ts	GET /api/map/route	Uses OSRM GIS engine, NOT an AI LLM model.
AI Fuel Optimization	■ MOCK / PLACEHOLDER	src/fleet/fleet.service.ts	GET /api/fleet/fuel	Manual fuel logs CRUD, no predictive ML model.
AI Driver Matching	■ MOCK / PLACEHOLDER	src/booking/booking.service.ts	POST /api/bookings/:id/accept	Broadcast dispatch + first-accept, no ML ranker.
Priority Logistics	■ FULLY IMPLEMENTED	src/booking/booking.service.ts	POST /api/bookings	Priority FCM alerts, surge rate, 15m delay monitoring.
Multi-Stop Waypoints	■ FULLY IMPLEMENTED	src/booking/booking.service.ts	POST /api/bookings	Sequence order, stop arrival, per-stop ePOD.
ePOD Verification	■ FULLY IMPLEMENTED	src/booking/booking.service.ts	POST /api/bookings/:id/epod	Stores digital signature & photo URLs.
Razorpay Integration	■ FULLY IMPLEMENTED	src/payments/payments.service.ts	POST /api/payments/create-order	HMAC SHA-256 signature verification, refunds.
Digital Wallet Ledger	■ FULLY IMPLEMENTED	src/wallet/wallet.service.ts	GET /api/wallet/balance	Double-entry ledger for deposits, escrow, payouts.
Driver GPS & Socket.IO	■ FULLY IMPLEMENTED	src/tracking/tracking.gateway.ts	Socket driver:locationUpdate	Live room broadcasting & stationary delay alerts.
FCM Notifications	■ FULLY IMPLEMENTED	src/notifications/...	POST /api/auth/push-token	Firebase Admin SDK multi-device messaging.
Fleet Asset Management	■ FULLY IMPLEMENTED	src/fleet/fleet.service.ts	GET /api/fleet/trucks	Vehicle, driver, maintenance, fuel, & tyre CRUD.

End of Official RouteX Technical Specification Document.